[INCH-POUND]
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

HONING MACHINES, HORIZONTAL, SINGLE-SPINDLE

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-H-80009B for all federal agencies.

1. SCOPE. This commercial item description (CID) covers single-spindle horizontal honing machines. The honing machines covered by this CID are intended for internal honing of open, blind-end, and interrupted bores generated by a previous forming or machining operation.

2. CLASSIFICATION. The honing machines shall be of the following types, styles, and sizes. The types, styles, and sizes to be furnished shall be as specified (see 7.2(b)).

Type I - Bench mounted
Type II - Floor mounted

Style A - Manual stroked
Style B - Power stroked

Size - Sizes are shown in table I

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3419

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TABLE I. Machine sizes.

Size	Bore diameter, inches	Bore length, inches	Type I		Type II	
			Style A	Style B	Style A	Style B
103	1	3	X	X		
205	2	5				X
312	3	12		X	X	
316	3	16				X
416	4	16				X
616	6	16			X	

Note: "X" denotes availability of sizes to type and style classifications.

3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The honing machine shall, as a minimum, include the following components:

3.2.1 Frame. The frame shall incorporate material, shape, and mass to minimize adverse vibration. The frame shall have provisions for leveling and anchoring the machine.

3.2.2 Work support. Style A manually stroked machines shall be equipped with workpiece support bars to absorb torque transmitted from the hone to the workpiece during manual stroking. Style B power-stroked machines shall be equipped with a workpiece support fixture to secure the workpiece during power stroking and shall be a universal fixture adaptable to a wide variety of workpiece configurations.

3.2.3 Splash guards. Unless otherwise specified (see 7.2(c)), the machine shall be equipped with splash guards considered standard by the manufacturer.

3.2.4 Spindle. All rotating parts within the spindle head shall be mounted on sealed, anti-friction bearings. The spindle drive mechanism shall provide variable spindle speeds necessary to achieve the performance requirements. The spindle shall have means for holding and driving all sizes of mandrels necessary to hone holes up to the maximum diameter of the machine size specified.

3.2.5 Power stroking mechanism. The power stroking mechanism on style B machines shall smoothly reciprocate the workpiece over the spindle without bind, vibration, or chatter.

3.2.6 Honing tools. If required, honing tools, honing stones, and other tooling shall be furnished as specified (see 7.2(d)). The honing tools shall be compatible with the stone expansion mechanism specified in section 3.2.7.

3.2.7 Stone expansion mechanism. Unless otherwise specified (see 7.2(e)), the machine shall have an automatic feed mechanism to expand and contract the honing stone(s) at a continuous, uniform, and preselected rate. The mechanism shall have provisions for manually adjusting the stone expansion rate and locking in the desired diameter setting.

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3.2.8 Coolant system. The machine shall have a coolant system consisting of a reservoir, a filter or sludge removal system, a pump, a coolant delivery nozzle, a flow control valve, and a pan for collecting spent coolant. The pump discharge capacity shall be adequate for honing parts within the specified size range of the machine and provide a full flow rate of clean coolant during the entire honing process. The system shall be equipped to allow the operator to direct and control the coolant flow to the workpiece.

3.2.9 Automatic size control. When specified (see 7.2(f)), the machine shall have an automatic size control system that stops the honing cycle when the workpiece bore reaches the preset diameter.

3.2.10 Square honing fixture. When specified (see 7.2(g)), the machine shall have a fixture to assure bore-to-face squareness of workpiece bores when the bore diameter is more than three times the length.

3.2.11 Operator controls. Unless otherwise specified (see 7.2(h)), the following operator controls and adjustments, as a minimum, shall be provided:

a. Power on/off switch(s)
b. Emergency stop switch
c. Foot pedal on/off and feed advance
d. Spindle speed
e. Feed rate
f. Coolant on/off switch or valve
g. Stroke length (style B only)

3.2.12 Electrical system. The electrical system shall conform to the American National Standards Institute/National Fire Protection Association (ANSI/NFPA) 79, "Electrical Standard for Industrial Machinery". Unless otherwise specified (see 7.2(i)), machines with motors of less than one horsepower shall operate from a 115-volt, single-phase, 60-hertz circuit; machines with motors of one horsepower and larger shall operate from a dual 230/460-volt, 3-phase, 60-hertz circuit and be wired initially for 230-volt operation. The control system shall operate on a 120-volt, single-phase, 60-hertz circuit.

3.2.13 Drive motor. The drive motor(s) shall be rated for continuous duty and have sealed and permanently lubricated ball or roller bearings. The drive motor(s) shall be energy efficient and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1, "Motors and Generators". Unless otherwise specified (see 7.2(j)), the drive motors shall have drip-proof enclosures.

3.2.14 Optional components. Optional components shall be provided as specified (see 7.2(k)).

3.3 Performance. Unless otherwise specified (see 7.2(l)), the machine shall meet the alignment requirements shown in table II and the performance requirements shown in table III when tested in accordance with section 5.3 through 5.3.3.3.

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TABLE II. Alignment requirements.

Spindle parameters	Tolerances
Runout, measured at spindle nose	0.001 inch, total indicator readout
End play	0.0002 inch
Spindle axis square to workholding fixture	0.001 inch per foot

TABLE III. Performance requirements.

Machine size	Maximum bore diameter, inches	Bore parameters			
		Maximum out-of-round[1], inch	Maximum straightness deviation, inch	Surface finish[2], microinches, R_a	Minimum metal removal rate, cubic inch per minute
103	2.0	0.0001	0.0002	5	0.05
205	2.0	0.0001	0.0002	5	0.05
312	2.0	0.0001	0.0002	5	0.05
316	2.0	0.0001	0.0002	5	0.05
416	4.0	0.0002	0.0002	5	0.50
616	4.0	0.0002	0.0002	5	0.50

[1] Measured at random intervals throughout bore length.
[2] Surface finish equal to or better than the value indicated. Surface finish, R_a, is the arithmetic average of the distance of the roughness profile from the mean line in accordance with ASME B46.1, "Surface Texture (Surface Roughness, Waviness, and Lay)".

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions as specified (see 7.2(m)).

3.5 Safety and health requirements. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with ANSI B11.9, "Safety Requirements for the Construction, Care, and Use of Grinding Machines", and Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "Occupational Safety and Health Standards for General Industry". If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(n)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(o)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

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3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(p)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer's name
c. Manufacturer's model designation and serial number
d. Power input (volts, total amperes, phase, frequency)
e. Short-circuit/over-current rating
f. Contract number or order number
g. National stock number
h. Date of manufacture

3.8 Lubrication plate or chart. When specified (see 7.2(q)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with Society of Automotive Engineers (SAE) number or lubricant identifier

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same machine offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The honing machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(r)), the honing machine shall be tested in accordance with sections 5.3.1 through 5.3.3.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(s)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the

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Government. The manufacturer shall provide test instruments and equipment calibrated in accordance with the applicable National Institute of Standards and Technology (NIST) standard. The manufacturer shall provide certification of calibration as specified (see 7.2(t)). Unless otherwise specified (see 7.2(u)), the machine shall be tested in accordance with environmental conditions specified in American Society of Mechanical Engineers (ASME) ASME B5.54, "Methods for Performance Evaluation of Computer Numerically Controlled Machining Centers", and ASME B89.6.2, "Temperature and Humidity Environment for Dimensional Measurement".

5.3.2 Operational test. The honing machine shall be operated in accordance with the manufacturer's standard operating procedures. The machine shall be operated at no load for not less than 60 minutes. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The honing machine shall be tested to determine compliance with the alignment accuracy requirements shown in table II and shall be subjected to the tests in sections 5.3.3.1 through 5.3.3.3. The choice of stones, coolant, stone feed rate, rotating and stroking speeds, and fixturing shall be the option of the supplier.

5.3.3.1 Test workpiece. The honing machines shall be tested using a section of American Iron and Steel Institute (AISI)/SAE 1026 cold-drawn seamless steel tubing. The workpiece shall have an inside diameter not less than 75 percent of the machine's maximum bore diameter and a length not less than 75 percent of the machine's maximum bore length. No pretest preparation of the workpiece shall be made to correct any inaccuracies that may be inherent to the workpiece as received from the mill.

5.3.3.2 Honing accuracy test. The test shall be performed using only one rough-honing operation and one finish-honing operation. The finished bore dimensions shall conform to the roundness, straightness, and surface finish requirements shown in table III.

5.3.3.3 Metal removal rate test. The metal removal rate shall be demonstrated on the workpiece used in the honing accuracy test in section 5.3.3.2. The metal removal rate shall conform to the requirements shown in table III.

5.3.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.2(v)).

5.4 Acceptance. Unless otherwise specified (see 7.2(w)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment accuracy tests shall be compared with the requirements shown in table II. The results of the performance tests shall be compared with the performance requirements listed shown in table III. Failure of the machine to meet the requirements shown in tables II and III shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the machine shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(x)).

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7. NOTES

7.1 Sources of documents.

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

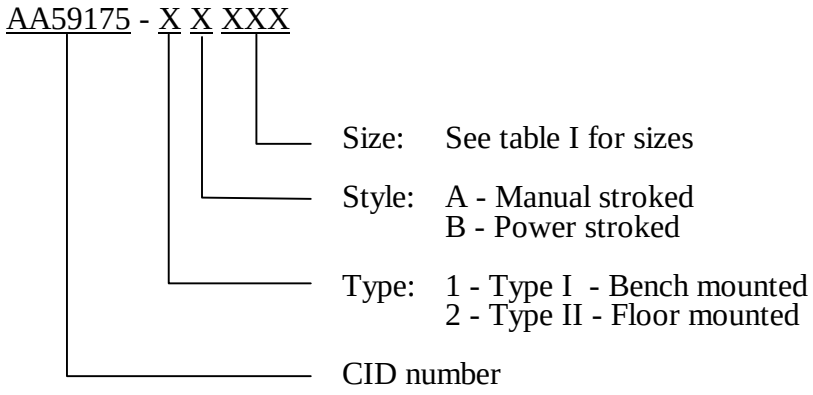
7.1.3 ASME standards. Copies of ASME standards may be obtained from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

7.1.4 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.5 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.6 OSHA standards. Copies of OSHA standards may be obtained from the Occupational Safety and Health Administration, U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



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7.3 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document
b. Size of honing machine required (see 2.)
c. Splash guards, if different (see 3.2.3)
d. Honing tools, stones, and other tooling, as specified (see 3.2.6)
e. Stone expansion mechanism, if different (see 3.2.7)
f. Automatic size control, if required (see 3.2.9)
g. Square honing fixture, if required (see 3.2.10)
h. Operator controls, if different (see 3.2.11)
i. Electrical system, if different (3.2.12)
j. Drip-proof enclosures, if different (see 3.2.13)
k. Optional components, as specified (see 3.2.14)
l. Performance, if different (see 3.3)
m. Machine dimensions and weight, if required (see 3.4)
n. System of units required (see 3.6)
o. Dual system of units, if required (see 3.6)
p. Nameplate, if different (see 3.7)
q. Lubrication plate or chart, if required (see 3.8)
r. Testing, if different (see 5.3)
s. Test responsibility, if different (see 5.3.1)
t. Certification of calibration, as specified (see 5.3.1)
u. Test environmental conditions, if different (see 5.3.1)
v. Optional tests, as specified (see 5.3.3.4)
w. Acceptance, if different (see 5.4)
x. Packaging requirements, as specified (see 6.1)

MILITARY INTERESTS:

Custodians:
Army - AL
Navy - SH
Air Force - 99

Reviewers:
Navy - AS, MC
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

(Project 3419-0233)

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